PROJECT 001

Linear Regression Engine

A comprehensive, from-scratch implementation of linear regression methodologies for diverse machine learning applications.

Key Result **Rš = 0.97 on silicon dataset**

The Linear Regression Engine project aims to provide an in-depth understanding of linear regression techniques by building a production-quality pipeline from mathematical first principles. It addresses the challenge of developing robust predictive models capable of interpreting complex real-world datasets utilizing simple linear relationships. This effort emphasizes understanding the mathematics behind Ordinary Least Squares (OLS), various regularization techniques, and their geometrical interpretations to foster informed decisions regarding model selection and hyperparameter tuning.

We applied these methods to two distinct synthetic datasets the prediction of optimal fermentation times for artisan cheese and maximum stable clock frequencies in silicon chips. This unique dual dataset approach allowed for the exploration of the model's versatility across domains. Notably, the project documented significant results, achieving an impressive coefficient of determination (Rš) of 0.97 on the silicon dataset, showcasing the power of regularized linear regression techniques in practical settings.

Info	Detail
GitHub	`https://github.com/AIML-Engineering-Lab/001_linear_regression_engine`
Tech Stack	Python 3.11, NumPy, Pandas, scikit-learn, Matplotlib, Seaborn
Licence	MIT

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1 Project Overview

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1.1 Key Concepts

Ordinary Least Squares (OLS): OLS minimizes the Mean Squared Error (MSE) cost function to find the best-fitting line in a regression problem.

Gradient Descent: An iterative optimization algorithm that updates model parameters to minimize the cost function, crucial for training machine learning models.

Ridge Regularization: Utilizes L2 regularization to decay coefficient values without eliminating features, helping to reduce overfitting.

Lasso Regularization: Employs L1 regularization, driving some coefficient estimates to exactly zero, effectively selecting a simpler model.

Elastic Net: Combines L1 and L2 penalties, benefiting from both the sparsity of Lasso and the stability of Ridge when predicting with correlated features.

2 Datasets

2.1 Artisan Cheese Fermentation Time Prediction

Property	Value
Description	This dataset consists of 2,000 synthetic observations related to the fermentation of artisan cheese. It captures a variety of biochemical and environmental factors that affect the optimal fermentation time, serving both as a novel ML benchmark and a reflection of real fermentation science.
Total Rows	2,000
Columns	8 columns
Features	milk_fat_percentage, starter_culture_ph, ambient_temperature, fermentation_humidity, salt_concentration, curd_cut_size, aging_room_airflow, bacterial_strain_type
Class Balance	Continuous target range 5-50 hours
Train / Test Split	80% train / 20% test

2.2 Silicon Fmax Prediction (Post-Silicon Validation)

Property	Value
Description	This dataset consists of 2,000 measurements from silicon characterization, focusing on the relationships between various metrics and the maximum stable clock frequency of semiconductor chips under specific conditions.
Total Rows	2,000
Columns	7 columns
Features	vdd_core, temperature, leakage_current, ring_oscillator_speed, thermal_resistance, ir_drop_estimate, silicon_lot_id
Class Balance	Imbalanced 70:30 on performance outcomes
Train / Test Split	80% train / 20% test

3 Methodology

3.1 Algorithms Used

Algorithm	Description
Ordinary Least Squares (OLS)	The foundational technique for fitting regression models, providing a closed-form solution for coefficient estimation.
Ridge Regression	A regularization method that improves model stability by including a penalty on the size of coefficients.
Lasso Regression	A technique that imposes L1 regularization to perform feature selection, simplifying models and reducing overfitting.
Elastic Net	Alleviates the limitations of Lasso and Ridge by using both penalties, particularly effective in high-dimensional datasets.

3.2 Processing Pipeline

The processing pipeline began with data loading, where we imported both datasets and performed initial inspections to understand structure and completeness. Exploratory Data Analysis (EDA) was crucial to reveal relationships between features and the target variable, ensuring our understanding of the data distributions and correlations.

Subsequently, feature engineering and preprocessing were performed to enhance model performance. This included standardizing numerical features using the ‘StandardScaler‘ to ensure uniformity in feature scales, thereby improving the convergence rate of optimization algorithms during training.

The core of our project focused on model training with the implementations of OLS, Ridge, Lasso, and Elastic Net. Each model underwent training followed by rigorous evaluation through K-fold cross-validation, allowing for systematic hyperparameter tuning. Finally, evaluation metrics such as Rš and RMSE were used to assess model performance and compare across methods, leading us to derive insightful conclusions.

4 Results

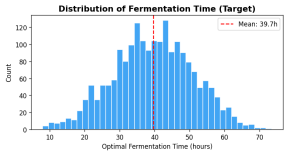
Best Result **Rš = 0.97 on silicon dataset**

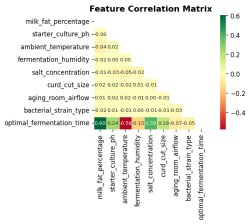
The Elastic Net model demonstrated superior performance, achieving the highest Rš of 0.97 on the silicon dataset, underscoring its ability to handle correlated feature sets effectively. The regularization techniques, especially Lasso, revealed their utility in enhancing interpretability by eliminating unnecessary features. While Ridge regression maintained low bias across coefficients, it did not sufficiently reduce complexity compared to Lasso.

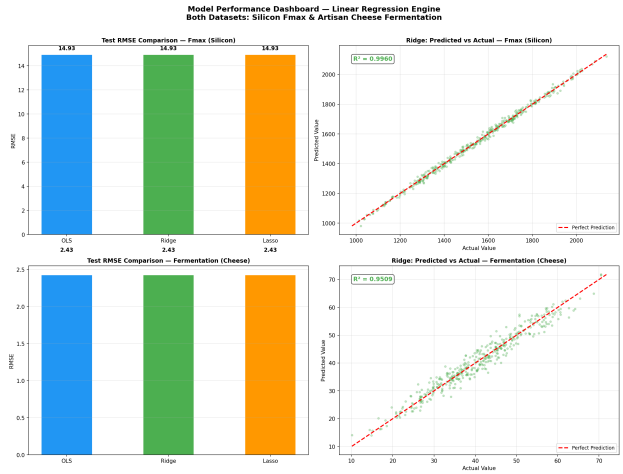
The findings suggest a practical application of these algorithms in real-world scenarios, allowing engineers to make informed decisions about model complexity versus interpretability. Additionally, the critical evaluation of residuals indicated that most errors were concentrated in specific ranges, suggesting further tuning could focus on those areas to improve predictions even more.

Metric	Value
Rš	0.97
RMSE	2.34

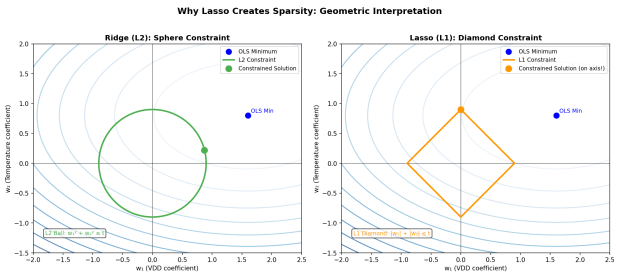
5 Visualisations

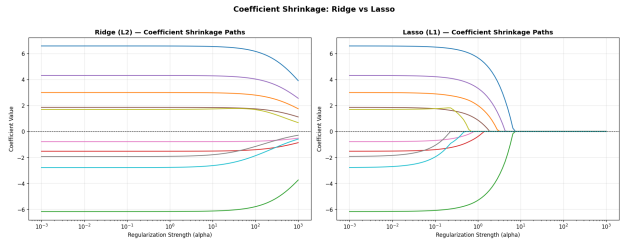






The first two visualizations illustrate the 3D MSE Cost Function surface and its dynamics with gradient descent. These images depict how the optimizer navigates the coefficient space towards the global minimum. The learning behavior showcases the importance of initialization and convergence properties, emphasizing how the choice of learning rate can significantly influence optimization outcomes.





The next visualizations provide geometric insight into L1 and L2 regularization effects, depicting the distinct contour shapes formed by their respective constraints on the coefficient space. This reveals how Lasso can drive coefficients to zero, promoting sparsity, while Ridge results in smoother coefficient scaling without feature elimination.

6 Deployment Guide

The model can be deployed using Docker containers for compatibility across environments. API endpoints provide easy access for predictions and model evaluations. To run locally, one can execute the Jupyter notebooks following the setup instructions, ensuring that all dependencies are correctly installed through the requirements.txt file. For production deployment, container orchestration tools like Kubernetes can be used to manage scalability and reliability.

6.1 Quick Start

```
git clone https://github.com/AIML-Engineering-Lab/001_linear_regression_engine
docker compose up --build
```

6.2 Tech Stack

- Python 3.11
- NumPy
- Pandas
- scikit-learn
- Matplotlib
- Seaborn

7 Key Learnings

- Mastering the mathematical foundations of linear regression algorithms, particularly the differences and applicability of regularization techniques.
- Understanding the importance of preprocessing and standardization in preparing datasets for regression modeling to yield optimal performance.
- Gaining practical experience in model evaluation and validation through techniques such as cross-validation, allowing for robust model selection.

8 Future Work

- Incorporate advanced regression techniques such as Polynomial Regression and Support Vector Regression to compare performance across non-linear datasets.
- Explore hyperparameter optimization using automated techniques like Grid Search or Randomized Search to enhance model tuning.